Media Richness Theory in Agile Development

Choosing Appropriate Kinds of Prototypes to Obtain Reliable Feedback

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***Abstract*—When product development involves many uncertainties and changes are highly likely, agile approaches are valuable. They integrate customers into the development process frequently and explore what satisfies them best. For that, the development team builds working prototypes, demonstrates them and receives feedback being incorporated into the next iteration. The team collects validated learnings and reduces risks early. However, building physical prototypes frequently is difficult, expensive and time-consuming. We contribute to that field by adapting the Media Richness Theory from communication research for agile development. As a result, we present a guiding model how to choose an appropriate kind of prototype depending on the complexity of communication task. In doing so, we illustrate it's plausibility with an exemplary case from practice.**

Keywords—Agile Development, Physical Product Development, Media Richness Theory, Kinds of Prototypes

I. Introduction

Typically, many unknowns exist in product development [1]. Some of them are known unknowns, others so called unknown unknowns. In contrast to unknown knowns, which might be less critical, these are main sources of uncertainty and volatility in product development and request the development team to learn. For that, agile development provides methodical frameworks like Scrum, Kanban or eXtreme Programming, which are well established in software development. One fundamental principle of agility is to build potentially shippable prototypes on a regular basis, so that the customer can feel and test them in order to give his valuable feedback to the team. By doing so, the team learns if the technical solution is feasible from an engineering point of view, desirable from the user's and customer's perspective, and viable from a business point of view [2]. Subsequent iterations, then, incorporate lessons learnt. In that way, the team reduces uncertainty early that means less changes in the future and more predictable project outcomes.

In contrast to pure software development, developing and building physical prototypes that are potentially shippable within a couple of weeks is much harder in most cases due to the constraints of physicality [3], [4]. Alternative kinds of prototypes like paper prototypes [5] or simulations [6] are cheaper and faster to build. However, they remain abstractions of reality and shrink the spectrum and amount of information being delivered to the customer. The more details the prototype incorporates, the higher is the chance to obtain reliable and, thus, valuable customer feedback to enhance the product realization. The paper at hand deals with this dilemma and investigates the appropriateness of different kinds of prototypes depending on the iteration's goal.

Limited to typed, verbal and body language the Media Richness Theory (MRT) explains how to choose an efficient communication medium in a person-to-person communication setting [7]. For the purpose of agile development of physical products, it might be possible to extent the MRT to object-based communication media. Thus, the research question reads as follows:

RQ: How can MRT give guidance to find appropriate kinds of prototypes in agile product development?

The goal of the investigation is a theory concept of how to combine the MRT with agile development in order to bring light in the selection of appropriate kinds of prototypes and, in turn, make agile development of physical products easier. For that, we argue and derive analogies between both research domains, but do not demand for sound empirical validation. An exemplary case study is utilized to explain the theory's plausibility.

We, firstly, give a brief introduction to the concept of agility, explain how it is linked to user experience and describe the original form of MRT. Secondly, we adopt the MRT to the context of agile development. A real case scenario, thirdly, exemplifies derived theory. Finally, we discuss both the theory and the example, and conclude the paper with limitations and remarks for future research.

II. State of the Art

A. *Agility, agile product development and the role of user experience*

In general, "[a]gility is the capability to react, and adopt to expected and unexpected changes within a dynamic environment constantly and quickly; and to use those changes (if possible) as an advantage." [8] Agile product development, consequently, brings that capability into the field of product development. It is very suitable for highly uncertain and volatile contexts in which initial circumstances and assumptions cannot be taken for granted. As agility does come at a cost, one should carefully evaluate if regarding context is unpredictable and volatile enough so that benefits exceed costs (compare e.g. commodities such as food or diapers versus long lasting, innovative development projects of complex technical systems such as production facilities).

Agile development originate from the software industry where the so-called manifesto of agile software development forms the basis of the practical implementation of above characteristics. The manifesto consists of four values and twelve principles that practitioners constituted for practitioners in 2001 [9]. According to the manifesto, a team working in an agile manner should be collocated, strongly committed to the project, and work tightly together on a daily basis and in a self-organized fashion. They should focus on customer satisfaction, thereby embrace changes, deliver working artifacts frequently, strive for technical excellence and prefer simple rather than complex solutions. Furthermore, developing in an agile way means to reflect the workflow honestly, openly, jointly and regularly. Agile methods such as Scrum, Design Thinking and eXtreme Programming concretize the way how the manifesto can be implemented in practice by suggesting practices, process routines and role definitions.

Scrum, for instance, gives implementation guidance by splitting the project in iterations (also called sprints) lasting 1 to 4 weeks [10]. They consist of analysing, planning, executing and reviewing activities [10]. Instead of analyzing customer requirements profoundly at the beginning, creating a plan and strictly executing it subsequently, it focuses on very small and manageable steps [10]. Consequently, it follows the Observe-Orient-Decide-Act loop [11] and fosters continuous learning. By presenting the customer what has been done in previous iteration, the customer gives comments, wishes and desires, which the teams considers in the next iteration [10]. Thus, each iteration can be perceived as experiment that tests uncertain assumptions with the intent to reduce risks frequently and early. Ideally, working prototypes are used for customer demonstration. The more reliable the customer feedback is, the better the team can learn how to achieve what satisfies the customer.

For that, user experience (UX) is key to ensure that the customer appropriately understands artifacts presented, and is able to interact with them as intended. Eason [12] points out that a large part of the potential of very useful technical systems remain hidden when users are only able to engage them to a certain extent. Thus, the question arises as to how a technical system has to be designed so that it is usable and useful for the user. The UX concept has its origin in software development and plays an increasingly decisive role in physical product development, too. Based on the standard for user-oriented design of user-friendly systems [13], one of the fundamental principles of usability engineering is the consideration of the entire user experience. While usability focuses on the actual usage situation and is defined as "the extent to which a product can be used by certain users in a particular usage context in order to achieve certain objectives effectively, efficiently and satisfactorily" [13], user experience is an extension to the anticipated use and processing of the usage situation [13], including all emotions, ideas, preferences, perceptions, physiological and psychological responses, behaviors, and achievements that result before, during, and after use [14]. Usability, as interpreted from the viewpoint of the user's personal goals, may include the type of perceptions and emotional aspects typically associated with the user experience. Criteria of usability can be used to assess aspects of the user experience. An early involvement of users in the design process as such, generally has a positive impact on user satisfaction, with the main factor resulting from the overall improvement in the quality of the requirements analysis [15].

B. *Communication, MRT, and related studies*

Communication is about sending and receiving intended meanings by using signs (visual, auditory, somesthetic, olfactory, gustative) and mutually shared interpretation rules (e.g. language). In the field of organizational information processing, the MRT explains how to choose an effective and efficient medium (face-to-face, telephone, letter etc.) to transmit intended meanings [7]. For that, it differentiates between media richness and communication task.

Media "[r]ichness is defined as the potential information carrying capacity of data. If the communication of an item of data, such as a wink, provides substantial new understanding, it would be considered rich. If the datum provides little understanding, it would be low in richness." [7] Media richness depends on the amount of cues (e.g. many and simultaneous irrelevant information), the response time, the customizability of the response, and the naturalness of the speech. According to Daft and Lengel [7], face-to-face has the highest richness and cannot be exceeded by other media.

The complexity of a communication task serves as measure of how easy it is to communicate a certain meaning. On the one hand, "[s]imple phenomena tend to be mechanical, routine, predictable and well understood. Simple phenomena mean that managers typically can follow an objective, computational procedure to resolve problems." [7] On the other hand, complex phenomena "are difficult, hard to analyze, perhaps emotion laden, and unpredictable. Managers have to spend time analyzing the situation and thinking about what to do." [7]

For the given complexity of a communication task, if the medium is too rich, communication might be inefficient (overcomplication). If the medium is too lean, misunderstandings due to a lack of information might be unavoidable (oversimplification). For example, telling someone that one will be late for a meeting (relatively simple communication task), would be overcomplicated by using a rich medium such as face-to-face, since too many cues are transmitted and an immediate response is not needed. Instead, a lean medium like a SMS would be an efficient choice. In contrast, giving someone in-depth feedback on his or her job (relatively complex communication task) via SMS or email would be oversimplified. Instead, a rich medium like a video call or face-to-face meeting would be sufficiently rich, but as lean as possible, to still be efficient and to avoid misunderstandings.

As an extension to the MRT, the Channel Expansion Theory (CET) developed by Carlson and Zmud [16], [17] argues that communication experience can compensate less rich media so that they become quasi-rich. Hence, media richness is not an objective characteristic of the particular medium [17], [18]. Rather, the medium's richness depends on the experience of the sender and the receiver whereas experience is four-dimensional [16], [17]: experience with the medium (e.g. how to use it), experience with the topic (e.g. which aspects are decisive), experience with the partner (e.g. preventing misunderstandings by knowing the partner's paradigms) and experience with the context (e.g. preventing misunderstandings by knowing the partner's circumstances and constraints).

C. *Prototypes and their role in agile product development*

The main reason for prototyping within agile product development is to reduce uncertainties as early as possible. According to [19], prototyping is part of an iterative learning process. He classifies the prototypes objectives into four main categories: exploration, communication, usability and design, and verification [19]. Möller [20] differentiates four prototyping categories within product development: concept models, geometrical prototypes, functional prototypes and technical prototypes. Spinuzzi [21] mentions that the creation of prototypes is an important tool for the discovery, evaluation and development of new product ideas. The creation of several prototypes during the development increases the understanding, specification and documentation of customer requirements. It enables the team to develop better results, that match the customer requirements to a greater extent [22]. Therefore, each prototype represents both, the knowledge and findings in a physical or virtual model [23]. Table 1 lists and describes some often-used kinds of prototypes.

TABLE I. DIFFERENT KINDS OF PROTOTYPES RANKED

Kinds of prototypes	Description
High fidelity mock-up	Detail-rich, often produced like the final product, looks and feels almost like the final product, can be tested in real environment, includes not necessarily all features of the final product
3D-printed model	Produced with additive manufacturing, focuses on only a few aspects like aesthetics or mechanical resilience
Laboratory set-up	Laboratory test, focuses on only few aspects like chemical processing, simulated environment
Simulation, CAD	Virtual prototype, simulated environment, often dynamic or interactive
Paper prototype	Made by paper, focuses on only few aspects like geometry
Lego prototype	Made by Lego, focuses on only a few aspects like system's architecture
2D-sketch or drawing	Hand-made or digital, technical or non-technical, focuses on only few aspects like geometry or assembly
Technical report	Verbal, may contain graphical illustrastions, focuses on only a few aspects like functional description
Calculation	Numeric, may contain some technical sketches

Media richness ↑

Prototypes in agile development are not only functionally reduced experimental models of later series products, but also deliver the confirmation of the question addressed to them as required [24]. They are simplified abstractions of the final product developed specifically for defined test purposes in the agile development process [25]. Prototypes are used for learning, communication, integration or demonstration [26]. They can take many different forms: from very rudimentary storytelling illustrations to paper and cardboard models, roll games to fully functional elaborations [27]. Thus, they can also be described as a "designer's multi-dimensional expression", meaning that designers can use models and prototypes to express their ideas in accurate and precise manners to others [28].

According to Brown [2], prototypes test feasibility (technology perspective), viability (business perspective) and desirability aspects (customer perspective) (see Fig. 1). As with Design Thinking, agile approaches iterate between user perspective (solution space) and product specification (problem space). The creation of numerous prototypes and the tests with customers and users enables new insights at an early stage [29]. According to Ries [30], a prototype is used to test a predefined business hypothesis. A prototype can be used to check the features and gather technical requirements, but also to optimize product properties [19]. Walker [25] distinguishes the prototype's fidelity, whereby a high-fidelity prototype is often made with the same methods as the final product. That is why it has the same interaction techniques and appearance as the final product but is more expensive and time-consuming to produce than a low-fidelity prototype [25].

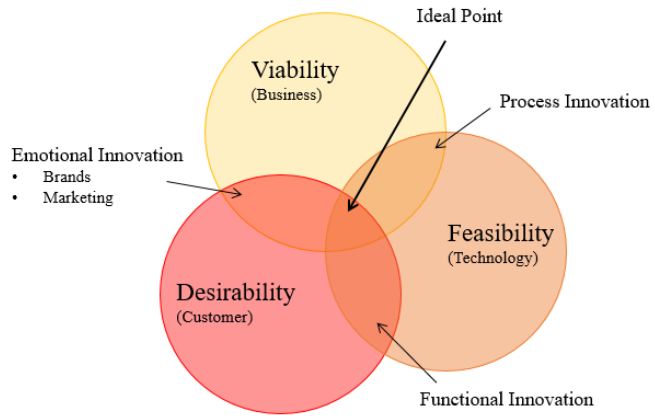


Fig. 1: Aspects of innovation by [2]

III. Media Richness Theory in Agile Development

A. *Setting adaption: MRT to MRT for agile development*

As practically shown in the field of agile development, prototypes can overcome language barriers between the different stakeholders [2]. Prototypes used as communication medium create a shared vision and a basis for further verbal discussion which has a positive impact on the project success [2], [31]. To make the original MRT applicable, we need to consider the changes of setting. The original MRT bases on a person-to-person as well as on a word- and behaviour-based language setting only. Now, for the context of agile development, the setting needs adaption since in agile development, the team presents their results (runs the experiment to learn and reduce uncertainty) by means of prototypes.

Firstly, since neither practice nor academia has figured out who is the most appropriate person reviewing the results at each iteration's end [32], [33], we call that role *'evaluator'* to remain neutral. The evaluator can be a user, sponsor, owner, maintainer, expert regarding some aspects, or internal or external surrogates of any before mentioned. Thus, the setting consists of the development team and the evaluator that communicate with each other (see Fig. 2).

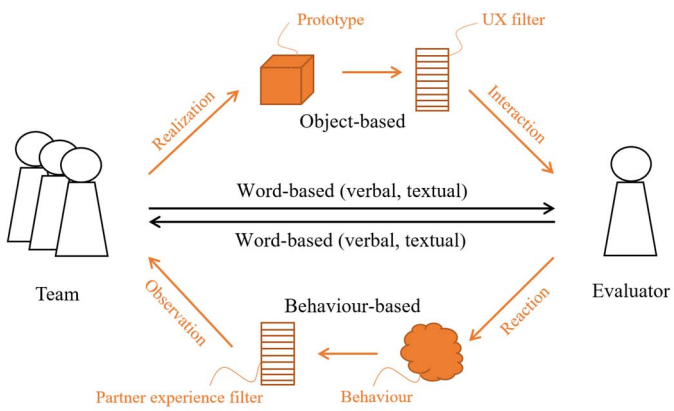


Fig. 2: Setting of adapted MRT

Secondly, the communication channels not only allow word- and behaviour-based information transfer, but also object-based demonstration. Originally, both parties communicate verbally (spoken or written) and via body language. In the adapted setting, the development team, on the one hand, uses mainly prototypes to send an intended meaning that needs to be interpreted by the evaluator. Depending on the experience and paradigm of the evaluator, information provided in form of the prototype can get lost. We call that user experience filter. To support proper understanding on the evaluator side the team can give additional verbal auxiliary explanation. On the other hand, the evaluator gives verbal feedback to the team (as in the original MRT), but also provides informative insights in the way he or she interacts with the prototype (behaviour). This, again, is filtered by partner experience referring to the team's experience with the user and his feedback.

B. *Theory adaption: MRT to MRT for agile development*

The MRT compares media richness to the complexity of communication task. For the former Daft and Lengel [7] rank common communication media according to their relative degree of richness (see Fig. 3). Face-to-face has the highest richness, followed by video and telephone conference, email and letter. Correspondingly, high fidelity mock-ups as close-to-final-product prototypes in at least some aspects stand for the richest kind of prototype. 3D printouts or laboratory set-ups typify similar richness as video conference, followed by virtual simulations (CAD, FEM etc.) and paper or Lego prototypes. 2D sketches, technical reports and numeric calculations on a sheet of paper represent the lower end of the richness rank for prototypes as they are far from the final product and use a rather abstract form of description being harder to interpret than the other ones. Media richness forms the y-axis (see Fig. 3).

On the one hand, rich kinds of prototypes such as high fidelity mock-ups stimulate many senses simultaneously and require ideally no auxiliary explanation by the team to support the prototype interaction. They have a high similarity to the final product in at least some aspects (such as one particular function) and tend to incorporate a great number of testable aspects. Rich kinds of prototypes generate the highest business value for the customer since they could potentially be used in the field as functional early product version. Consequently, rich kinds of prototypes provide a large variety of information concerning the prototype that – depending on the complexity of communication task – could distract the evaluator from the intended question to be tested. On the other hand, lean kinds of prototypes such as sketches or numeric calculations are easier to modify spontaneously by the development team and allow quick experiment changes that prevent the team from postponing the changed experiment to the next iteration (also compare rehearsability proposed by Dennis and Valacich [34]). Thus, they provide flexibility and a higher degree of freedom, but transfer only less information.

Similar to the original form of the MRT, the x-axis incorporates the complexity of communication task. In agile development, a communication task summarizes the experiment to be run in a certain iteration that questions a particular engineering design uncertainty. Simple communication tasks tend to be closed questions referring to experiments with a fixed,

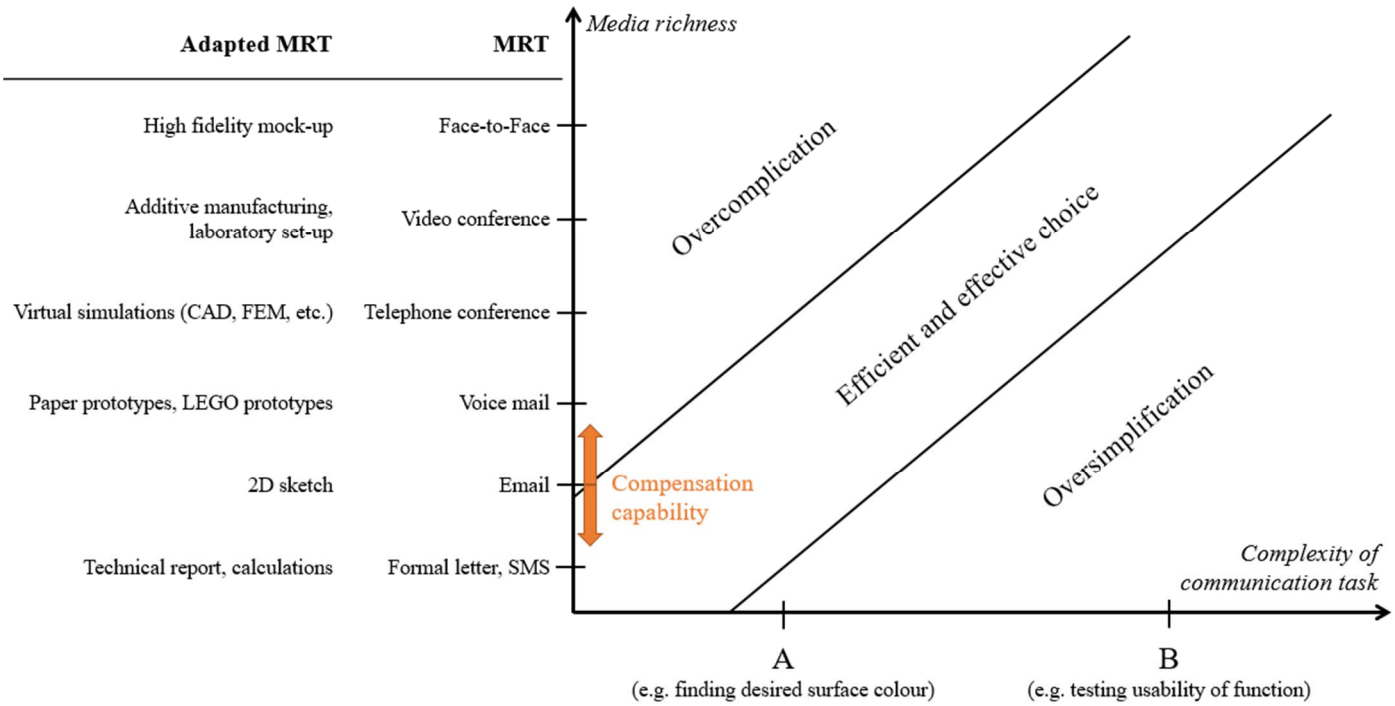


Fig. 3: Adapted MRT for agile development

often pre-known set of answers rather binary in nature. Finding an answer to the corresponding question is easy; methods are well known, computational and objective. In Fig. 3, example A points to a closed set of colour options that could be tested, for instance, in coloured product sketches which the evaluator can choose from. Complex communication tasks (compare example B), however, refer to open questions with unlimited answer options. The topic to be investigated is difficult to oversee, as many factors influence the answer or the relationships between the factors are hard to understand. The more complex the task, the richer the communication medium should be in order to provide required information and to prevent misunderstandings. This leads to the efficient and effective trade-off area.

Oversimplification (too lean) comes along with misunderstandings due to missing information. Overcomplication (too rich) generates ambiguity as too many irrelevant information are provided that can confuse or distract from the intended meaning. The intermediate trade-off area assigns appropriate (meaning efficient and effective) choices of prototype kinds depending on the complexity of the communication task. Appropriate choices are characterised by providing just the right amount of relevant information. In other words: The kind of prototype is appropriate for corresponding communication task, if it is rich enough to deliver needed information to establish sound understanding, but simultaneously lean enough to avoid ambiguity. Selecting an appropriate kind of prototype maximizes the chance that the intended meaning gets fully understood by the evaluator which, in turn, maximizes the chance to obtain reliable and, consequently, valuable feedback.

According to the CET, compensation capacity does exist which depends on the experience with the medium, topic, communication partner and context [16], [17]. The former refers to the familiarity with a certain kind of prototype. The more often and intense a certain kind has been used, the more design facets can be transferred and understood properly. Experience with the topic is about the knowledge both parties have in the particular subject (e.g. aesthetics of cars). The more often and intense the knowledge, the more involved persons know where to focus on [35]. Experience with communication partner allows to highlight aspects that are – out of experience – at risk to be misunderstood. Finally, experience with the context deals with the empathy and knowledge about the circumstances and constraints one party has for the other one. Consequently, the compensation capacity is bilateral, composed by the experiences of both parties the team and the evaluator. As an easy example, mechanical engineers have a trained spatial sense allowing 2D drawings or sketches to be easily imaginable in 3D. However, it might not be possible to fully compensate the richness of a high-fidelity mock-up.

A key motivation of agile development is to build artefacts, measure their impact when introduced to the environment and learn or rather evaluate the interaction [36]. However, it is difficult to delineate the term evaluation due to its complexity and variety of application areas. Evaluation may be internal or external, quantitative or qualitative, or a mix of these. Taking into account not only on viability, but also on attributes like feasibility and desirability challenges the objectivity, comparability and traceability of any evaluation. Analogously to the concept of product/market fit (a product should satisfy an essential demand in a certain market to a large extent to make the development and launch endeavor successful on the long run), there is a fit between communication and prototype. Fit means the prototype needs to be accurately adapted to the intended communication referring to the trade-off area in Fig. 3 in order to obtain reliable stakeholder feedback.

IV. Illustrative Example

The following outlines an illustrative example by using the case of LumiNav. We apply the theory ex-post to that case in order to test the validity and the value of the model. The team was part of the interdisciplinary course Think.Make.Start. at the Technical University of Munich in cooperation with UnternehmerTUM. The team`s idea was to build a smart indicator integrated in the sides of a handlebar showing you the navigation directions (compare also Fig. 4). The team consisted of five team members with majors in business, mechanical engineering, electrical engineering and computer science, representing every competence needed to build a LumiNav.

On the first day, the team started to build simple prototypes. Through a survey of about 30 people of different ages and lifestyles, they learned that LumiNav needed to be a removable all-in-one device sitting in the middle of the handlebar. Apart from these design choices, the team learned that the basic features needed on a bike are a bright light, a navigation function and a trip computer including a speedometer (and additional features like a health tracking system). The speedometer is needed because it is not allowed to drive faster than 30 km/h on Munich´s bicycle streets by law – fines can be the consequence.

For the first two prototypes, the team focused on design prototypes to explore the desirability of the design with potential user. The first two versions do not only represent a different design, but also incorporate two objectives with a simple demonstration of functionality. The first prototype already had a working light, whereas the second one simulated a working navigation system by the use of post-it`s. Since the faked navigation system is a little bit harder to communicate than a light, the second prototype is slightly more complex concerning the communication task.

The results of the surveys led the team to a combined slim design with curvy outlines, which they received really good feedback for. Based on that, the team focused on feasibility and design by the use of a CAD program. Modelling the casing with CATIA, they virtually arranged the approved features of the prototype. The complexity of the communication tasks within the team was simple but the media richness kind of high. Since a CAD model has a higher degree of fidelity than a paper prototype, Prototype #3 is arranged higher than prototype #1 and #2. Subsequent to an internal team discussion, the casing passed through another design review to become more slimmer in width (compare prototype #4).

In this design, the team integrated a front light and a combined LED distance control to prove via surveys that the concept is intuitive and the functionality is adequate. Therefore, prototype #4 focused on desirability of the prototype`s functionality (LED distance control). The team particularly used the prototype to communicate the "new" way of LED distance control. The nearer one comes to the intersection on which a turn is necessary, the more LEDs are shining.

After that, they started modelling a product structure and a morphological box, which assisted them to decide which technology is good to use. Through these processes, the team got a better focus on the main features helping to start faster developing prototypes that are more advanced. Prototype #5 was mainly focusing on feasibility and technical verification. The functional prototype was embedded in a laser cut housing, since the fidelity of the electronic components was not high enough yet, for the desired casing.

With prototype #6, the team also focused on feasibility, but also on usability and design, as well as verification. The team tried to find out which available rapid prototyping machine was adequate concerning manufacturing of a small production series.

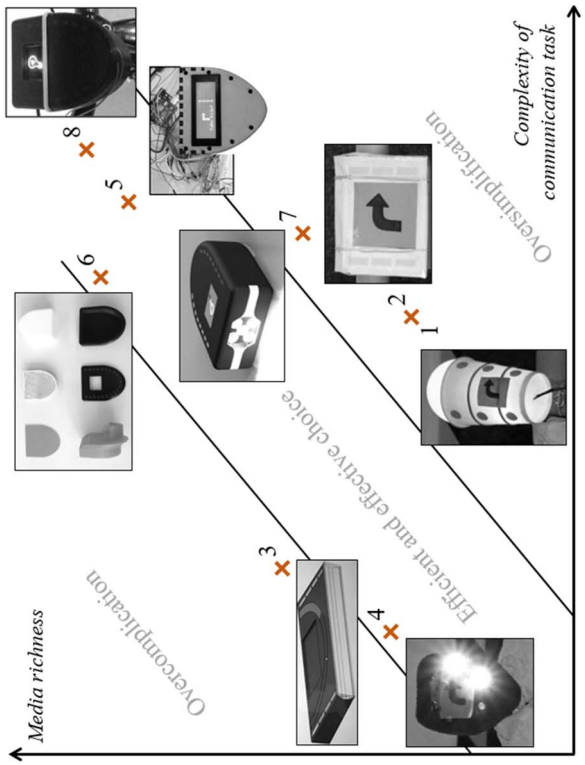
The CAD rendering, prototype #7, already pictures the final concept with regards to high fidelity. It combines a high-resolution display, a bike light approved by law and a discrete designed chain of LED lights, helping the user to orientate intuitively to the right direction depending on the current distance. The aim of this virtual prototype was to check feasibility, design and verification aspects. In addition, the prototype was used for the business plan. In parallel to the development, the team switched from B2C to B2B market. In contrast to the B2C market, the B2B business model seemed more promising, especially for the first step into market. This plan was confirmed through an interview with the owner of a bike rental shop, who stated that he would like to test the device. He gave some constructive feedback about adding preinstalled tours for tourists and a discount system at local sightseeing locations, combined with smart advertising.

The final prototype can be seen as a Minimum Viable Product with minimal faults. The rubber seen in the picture, for instance, holds the housing as the students simply forgot a mounting and a battery charging solution. The main objective of the last prototype is to test the viability of the entire system, and therefore to communicate the product as a whole. Additionally, the prototype includes a final verification of the technical concept.

Fig. 4 gives an overview of the prototyping roadmap and the table specifies the purpose for which the prototypes respectively were created as well as what kind of prototype was built. The numbers indicate the chronological order. As an implication to the ex-post theory application to the LumiNav case, we found that the team chose quite effective and efficient kinds of prototpyes depending on the complexity of the communication task.

The team successfully managed to overcome the constraints of physicality by using simple and easy-to-build kinds of prototypes that still transfer a sufficient spectrum and amount of information to the evaluator. However, prototpyes #1, #2, #3 and #7 leave potential for improvement. Prototype #3 (CAD model), for instance, is too rich to test wether the electrical components fit into the casing. Another paper prototype would have been easier and cheaper to build, to communicate and to be evaluated by the customer.

The case proves that it is possible to build physical prototypes in an agile manner within a couple of days. High fidelity mock-ups are not always the perfect means of choice, although it is common practice in German industry. For future development projects, the derived theory can give guidance to optimize the selection of appropriate kinds of prototypes. It makes agile development of physical products easier as it enables the team to build prototypes in very short iterations.



#	Uncertainty	Hypothesis
1	What is the critical or most desirable product feature?	User wants an „All-in-One“ device for the own bike (including: bike light, navigation, and extra features).
2	What is the critical or most desirable product feature?	User wants an „All-in-One“ device for the own bike (including: navigation, and extra features).
3	Do the electrical components fit into the casing?	Electrical components do fit into the casing.
4	Does the customer wants navigation indicated by lights?	User wants small device that shows navigation direction and gives orientation by light signs.
5	Are we able to build a working product system?	Team is able to build a working prototype.
6	Which additive manufacturing procedure is appropriate for our system's casing?	3D printing is a good way to manufacture casing, but different printers have various characteristics.
7	How does the product look like as a whole?	User wants an „All-in-One“ device for the own bike (including: bike light, navigation, and extra features)
8	Is our system's design as a whole viable for our sponsor?	User wants an „All-in-One“ device for the own bike (including: bike light, navigation, and extra features).

#	Prototype description	Category	Kind of prototype used	Type of prototype used	Purpose	Evaluator
1	Representing bike light and navigation display on top of handlebar.	Desirability	Paper prototype	Design prototype	Exploration	Potential user
2	Representing small navigation display with several use cases on top of handlebar.	Desirability	Paper prototype	Design prototype	Exploration	Potential user
3	Representing bike light and navigation display on top of handlebar.	Feasibility	CAD model	Geometric prototype	Design	Team
4	Representing bike light and navigation display on top of handlebar.	Desirability (feasibility)	Styrofoam model	Functional prototype	Exploration	Potential user
5	Working display with navigation and integrated orientation lights	Feasibility	Laboratory set-up	Functional prototype	Verification	Team
6	Casing manufactured by the use of different 3D-printers.	Feasibility	Additive manufacturing	Functional prototype	Usability, design, verification	Team, (potential user)
7	Virtual version of final concept with all features integrated.	Feasibility	CAD model	Design prototype	Usability, design, verification	Team
8	Final concept with all features integrated, manufactured components.	Viability	High fidelity mock-up	Technical prototype	Validation, communication	Potential sponsor

Fig. 4. Prototypes built by the team of students to design LumiNav. Placement in the adapted MRT theory (top left), uncertainty tested (bottom left), and further prototype descriptions (right).

V. Discussion

A. Discussion of the model

By definition, product development is about deriving a "recipe" of (a) how a product should be in order to satisfy customer needs and of (b) how to produce it [37]. "Perhaps the most important aspect of the recipe is confidence in it's ability to satisfy project stakeholder, who desire certain levels of product performance (which includes a product's features, functions, aesthetics, reliability, producibility, cost, etc.)." [37] A lack of confidence can stem from unclear needs, less known or mastered technology, rising product complexity, instable equilibria in competitive environments and evolving organizational structures of companies.

For that, Sutcliffe and Sawyer [1] utilize the classification of known knowns, unknown knowns, known unknowns and

unknown unknowns. In MRT research, Daft and Lengel [7], [38] propose a differentiation between uncertainty and equivocality. The former deals about the lack of information, whereas the latter refers to situations in which information exist but the interpretation is unclear. This differentiation would be beneficial in product development research as well, since both, information and its interpretation, are needed to gain confidence.

Agile development is a conducive approach to continuously learn and reach the required level of confidence quickly. Presented theory helps to bring light in how to transfer agile development from software industry to the development of physical products as adopting well-proven agile practices from software development is not trivial in physical product development. The theory presented argues that it is not necessary to build prototypes that reflect every function in a maybe early version of the final product. Product developers in a highly uncertain and volatile environment should rather build prototypes that focus on only one or a few design aspects. The overarching goal should be validated learning what satisfies the customer best. By building many small and simple prototypes, the development team can reduce risks early. Depending on how difficult it is to communicate information required for evaluation, prototypes do need to reflect neither a large variety of function nor a high similarity to the final product. Furthermore, the customer is not in all cases the most reliable source for prototype evaluation. Depending on the topic, especially when it comes to uncertainty regarding viability or feasibility, other internal or external roles might give a more reliable feedback than the customer might, although customer integration is important for desirability in particular.

However, presented theory is not suitable for every setting. It assumes that it is purposive to bring the information about the need (what satisfies the customer best) carried by the evaluator to the development team. Thereby the development team carries the knowledge about the technical realization and merges these two constituents during the product development process. However, Kruchten [39] reports that in some cases (e.g. food development) it is cheaper to bring the technological knowledge to the evaluator than the need information to the development team. Then, presented theory is not applicable since the development team provides tools that enable the evaluator to find the best way to satisfy the need. Corresponding communication setting is different to the original MRT setting and to the one derived here.

The existence of compensation capacity has an important implication. Internal customer surrogates, which is an often-used practice in industry currently [33], are beneficial in the way that they lower the effective and efficient trade-off area and keep the prototypes cheap and fast to build. This is because surrogates (e.g. engineers) have experience with the kind of prototype (medium), product (topic), the development team (partner), and the user context as well as the development team context (context) and can, thus, compensate missing media richness. It increases flexibility in product development [22] since leaner kinds of prototypes become effective and efficient, but risks that the surrogate is not in line with the exact customer desires.

Although Fig. 2 illustrates the object-based communication to be unidirectional from the team to the evaluator, also the evaluator can use object-based communication by pinpointing particular characteristics of presented prototypes when giving feedback. To a certain degree, the evaluator can draw quick sketches that as well can serve as basis for the feedback. It reduces ambiguity and fosters the creation of a shared product vision between the evaluator and the team. This is called reprocessability in Media Synchronicity Theory by Dennis and Valacich [34] and has the potential to further extend the application of the theory in agile development.

Another factor in Media Synchronicity Theory is rehearsability that refers to the ability to spontaneously revise or modify the message by the sender [34]. In object-based communication in agile development, rehearsability is more difficult and time-consuming than in word- or behaviour-based communication, yet powerful for learning purposes. However, especially lean kinds of prototypes such as sketches, calculations and reports might be reasonably quick to revise. This, again, implies potential for further theory extension.

In particular, companies enjoying and relying heavily on good brand reputation might choose appropriate kinds of prototypes differently, especially when it comes to prototype demonstrations to external evaluators (external to the company). They communicate less openly because they fear to lose brand reputation when presenting unfinished product versions or artefacts. Prototypes represent the brand, and customers expect them to have high quality according to the brand reputation. If they do not measure up to the customer's expectations, their brand reputation might be harmed on the short run. On the long run, the brand reputation benefits from frequent customer or user integration into product development and increasing product quality. Besides using internal surrogate of external stakeholder, a target group classification of prototype presentation according to Jensen et al. [40] could help to find compromises. Jensen et al. [40] differ between internal/internal (to communicate within the development team), external/internal (to communicate with company employees not being part of the development team) and external/external prototypes (to communicate with stakeholders not being part of the company).

According to agile methods such as Scrum, prototypes shall be product increments that are potentially shippable. In physical product development, this is often impossible to be accomplished within 1 to 4 week iterations. To tackle that, development teams could enlarge the cycle duration to a couple of month which influences the ability to react on expected and unexpected changes negatively or alternatively implement an discontinuous development cycle as proposed by [41]. Based on the presented theory we suggest development teams to reinterpret the purpose of iterations in agile development of physical products. Instead of striving for a product increment that represent a part of the final product, the team should perceive iterations as experiments to reduce uncertainty and to test assumptions. In this way, the team does not build modules that accumulate to the final product in the end. Rather, the development team collects validated learnings about the product requirements as well as admissible technological and technical realization possibilities that consolidate a certain design.

For that, we recommend following steps that base on the logic illustrated in Fig. 5:

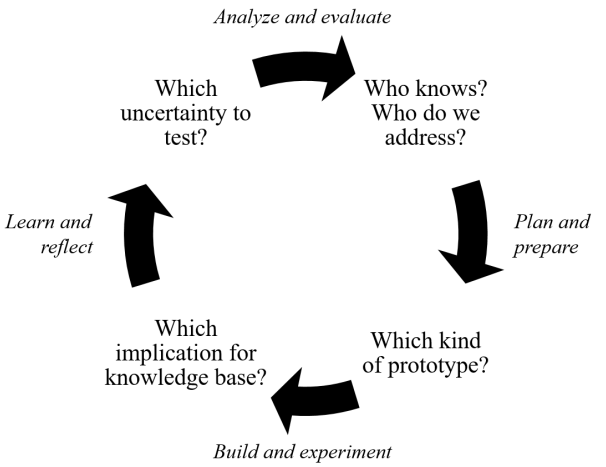


Fig. 5. Recommended logic of how to reduce uncertainty in product development by means of prototypes.

1. The team should ask themselves which uncertainties exist and which one is currently most crucial to reduce.
2. The team should formulate a corresponding question of investigation that will dominate the upcoming iteration. Every action that contribute to that question is highly welcome. However, the results of pursued action should be on hand, at latest when the iteration ends.
3. The team should ask themselves who can answer the question most reliably. Depending on the question and the product, it may be internals or externals. This influences the choice of prototype kinds and the necessity of customer integration.
4. The team should ask themselves how difficult it is to communicate the information, which the evaluator needs for answering the question most reliably (complexity of communication task).
5. By using the adapted MRT theory for agile development, the team can find an appropriate kind of prototype within the trade-off area. However, the team should consider that compensation capacity, reprocessability, rehearsability and brand reputation might influence the choice, too.
6. The team then should design and run the experiment that tests the questions.
7. The team should rerun these steps for each iteration.

An important issue remains unsolved: How can a team know that it tests the right aspect in current iteration? Maybe there are hidden unknown unknowns that will pop up and call previous development results into question. This is in the nature of product development. We suggest to collect sources of uncertainty jointly, to prioritize them based on the team's experience and to start testing the most crucial one first. By immediately going ahead and trying things out the team learns how to achieve the goal quickly.

Nevertheless, the risk is that the bits and pieces of knowledge collected during the experiments do not accumulate to the final product. In that way, the development team learns but does not generate customer value, which leads to a failed project. It requires a strong team commitment to the development task and team desire to deliver. Instead of establishing it through extrinsic motivation, intrinsic motivation might be more effective and efficient. This can be achieved by close customer collaboration and by fostering the understanding of incitements that cause customer desires and wishes.

B. *Discussion of the illustrative example*

As shown in the exemplary theory application, presented theory seems to be plausible in practice. The scatter plot spreads diagonally in Fig. 4 and follows the diagonal trade-off area. Thus, all prototypes either lie within that area or are close to it. Nevertheless, strong empirical evidences are missing for sound theory validation. It calls for further research to gain a high level of confidence.

Two outliers exist which implies improvement potential for the development team investigated. In the first two iterations, they face high, crucial uncertainty in a topic that is relatively complex to communicate and test. They build paper prototypes representing moderate richness that is not sufficient according to the theory. The intended evaluator is the user who has had little experience with that kind of prototype. Hence, compensation capacity cannot explain the gap between the chosen kind of prototype and the one suggested by the theory. However, as proposed in design thinking quick prototyping that tests design assumptions in very short iteration cycles is promising [2]. Then, high reprocessability and rehearsability is more important than appropriately rich prototypes that perhaps manifest a design idea too rigidly and suppress constructive and creative feedback, especially in early phases [2].

VI. Conclusion

Building physical prototypes frequently, which agile development fundamentals calls for, is difficult, expensive and time-consuming. Choosing the prototypes to use, and deciding when and how to carry those out not only depend on the nature of the product being developed, but mainly on the set of skills that the development team and the evaluators possesses in order to understand them. The aim of the paper at hand was to contribute to that field by adapting the MRT from communication research for agile development. Therefore, we outlined an illustrative example by using the case of LumiNav and, as a result, presented a guiding model how to choose an appropriate kind of prototype depending on the complexity of communication task.

Based on the presented theory we, therefore, suggest development teams to reinterpret the purpose of iterations in agile development of physical products. Instead of striving for a product increment that represent a part of the final product, the team should perceive iterations as experiments to reduce uncertainty and to test assumptions. In this way, the development team collects validated learnings about the product requirements as well as admissible technological and technical realization possibilities that consolidate a certain design.

Especially lean kinds of prototypes such as sketches, calculations and reports might be reasonably quick to revise. This, again, implies potential for further theory extension. Additionally, in our use case, it became clear that the MRT's contributory value for agile development differs along respective usage contexts and settings that might be worth investigated. Finally, the theory presented is a concept derived from theoretical arguments. Further empirical studies need to be pursued to empirically prove and validate the theory, whereas the first application in the case of LumiNav seems promising.

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